Our speed...

Let us play a small game. Go to the playground with your friends. Mark 100 metre distance for a race. Conduct a friendly running race and calculate the time taken by them to complete the distance. Now record the time in the table.

S. No	Name of the Student	Distance	Time taken (in seconds)	Speed = $\frac{\text{Distance travelled}}{\text{Time taken}}$	Speed (m/s)
1	Murugesan	100 m	12 S	100 M / 12 S	8.3 m/s
2		100 m			
3		100 m			
4		100 m			
5		100 m			

If you know the speed of an object and the time taken by it, then we can compute how much distance it had travelled.

We know that,

$$\text{Speed} = \frac{\text{Distance travelled}}{\text{Time taken}}$$

$$s = \frac{d}{t} \text{ or } st = d$$

Therefore, the distance travelled (d) = speed (s) × time (t).

$$d = s \times t$$



Usain Bolt crossed 100 metre in 9.58 seconds and made a world record. If you are able to run faster than him, then Olympic Gold Medal is waiting for you.

If a ship travelled at a speed of 50 kmph and it sailed for five hours, how much distance it has travelled?

Distance = s × t

= 50 kmph × 5 h = 250 km

If we know the speed and distance travelled we can compute the time taken.

$$s = \frac{d}{t} \quad \text{or} \quad t = \frac{d}{s}$$

Time taken = Distance travelled / Speed

Suppose a bus travels at a speed of 50 kmph and has to cover a distance of 300 km, how much time will it take?

$$t = \frac{d}{s} = 300 \text{ km}/50 \text{ kmph} = 6 \text{ h}.$$

Compute the following Numerical Problems.

1. If you travel 10 kilometres in 2 hours, your speed is _____ km per hour.
2. If you travel 15 kilometres in 1/2 hour, you would travel _____ km in one hour, and your speed is _____ km per hour.
3. If you run fast at 20 kilometres per hour for 2 hours, you will cover _____ km